TASK 8: Simple Sales Dashboard (Python Version)

Dataset + Code + Visual Output

Generated on: 2025-08-17 06:37:06

Rows: 4,599 | Date range: 2024-01-01 to 2025-07-31

Columns: Order ID, Order Date, Region, Category, Sales, Profit, Month-Year

This report includes:

- Dataset preview
- Full Python code listing
- Line chart: Sales over Months
- Bar chart: Sales by Region
- Donut chart: Sales by Category
- 3–4 insights

Dataset Preview (first 25 rows)

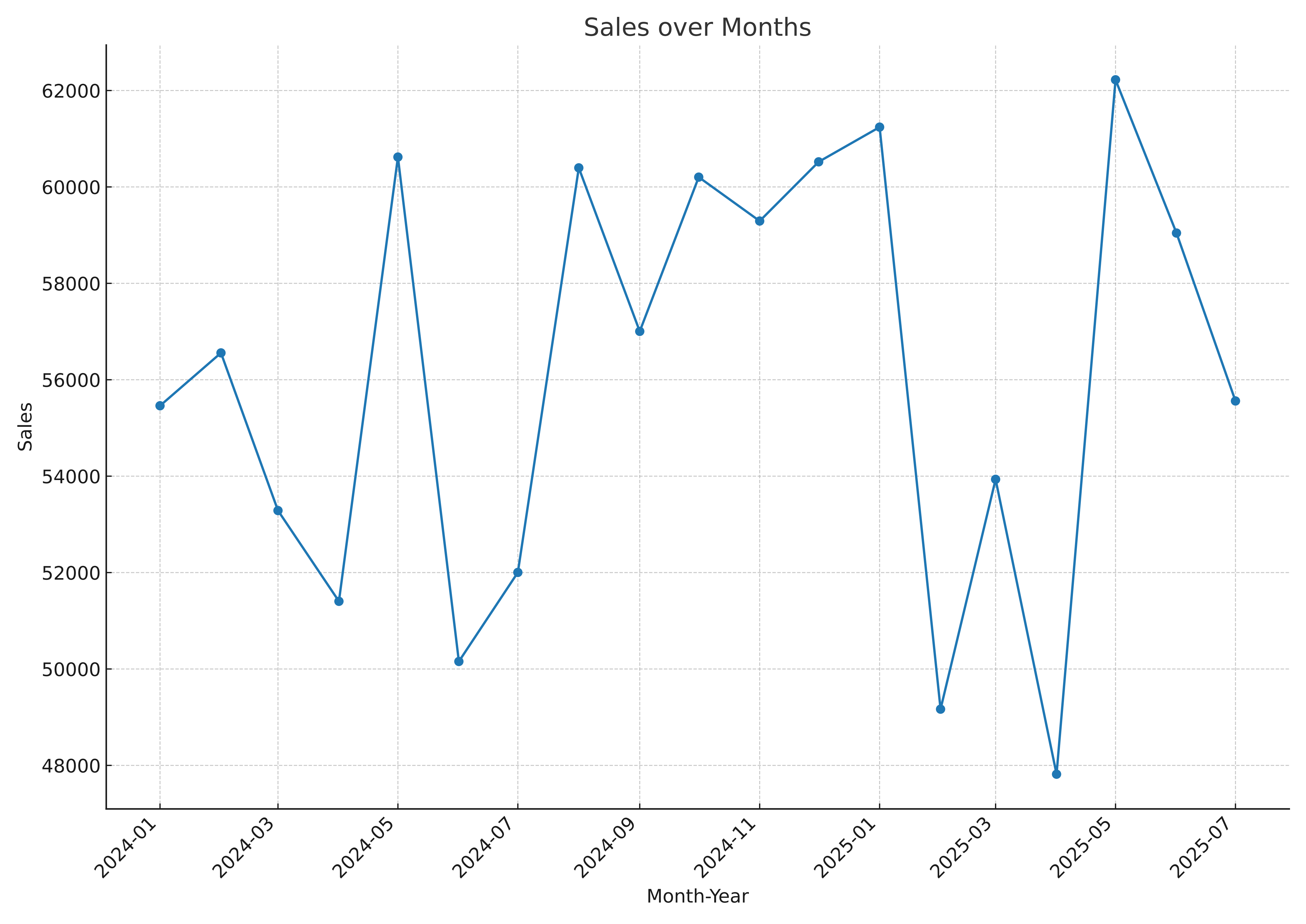
Order ID	Order Date	Region	Category	Sales	Profit	Month-Year
1	2024-01-01	South	Office Supplies	95.60	14.70	2024-01
2	2024-01-01	East	Furniture	21.40	4.87	2024-01
3	2024-01-01	West	Office Supplies	102.02	14.58	2024-01
4	2024-01-01	West	Office Supplies	93.56	19.64	2024-01
5	2024-01-01	West	Office Supplies	125.13	19.30	2024-01
6	2024-01-01	East	Furniture	197.48	44.90	2024-01
7	2024-01-02	South	Furniture	139.30	32.68	2024-01
8	2024-01-02	West	Office Supplies	129.18	28.95	2024-01
9	2024-01-02	South	Furniture	161.71	16.31	2024-01
10	2024-01-02	West	Technology	211.47	28.75	2024-01
11	2024-01-02	West	Furniture	300.94	67.53	2024-01
12	2024-01-03	East	Furniture	353.23	47.72	2024-01
13	2024-01-03	Central	Technology	254.70	44.88	2024-01
14	2024-01-03	Central	Technology	609.36	119.99	2024-01
15	2024-01-03	West	Office Supplies	103.26	18.27	2024-01
16	2024-01-03	East	Furniture	125.86	29.22	2024-01
17	2024-01-03	West	Technology	325.18	34.22	2024-01
18	2024-01-03	Central	Furniture	231.44	36.72	2024-01
19	2024-01-04	East	Office Supplies	132.44	25.57	2024-01
20	2024-01-04	East	Office Supplies	60.55	9.63	2024-01
21	2024-01-04	East	Furniture	235.89	47.23	2024-01
22	2024-01-04	West	Office Supplies	130.82	23.06	2024-01
23	2024-01-04	South	Furniture	82.11	14.67	2024-01
24	2024-01-04	West	Furniture	255.27	77.39	2024-01
25	2024-01-04	East	Furniture	233.17	45.49	2024-01

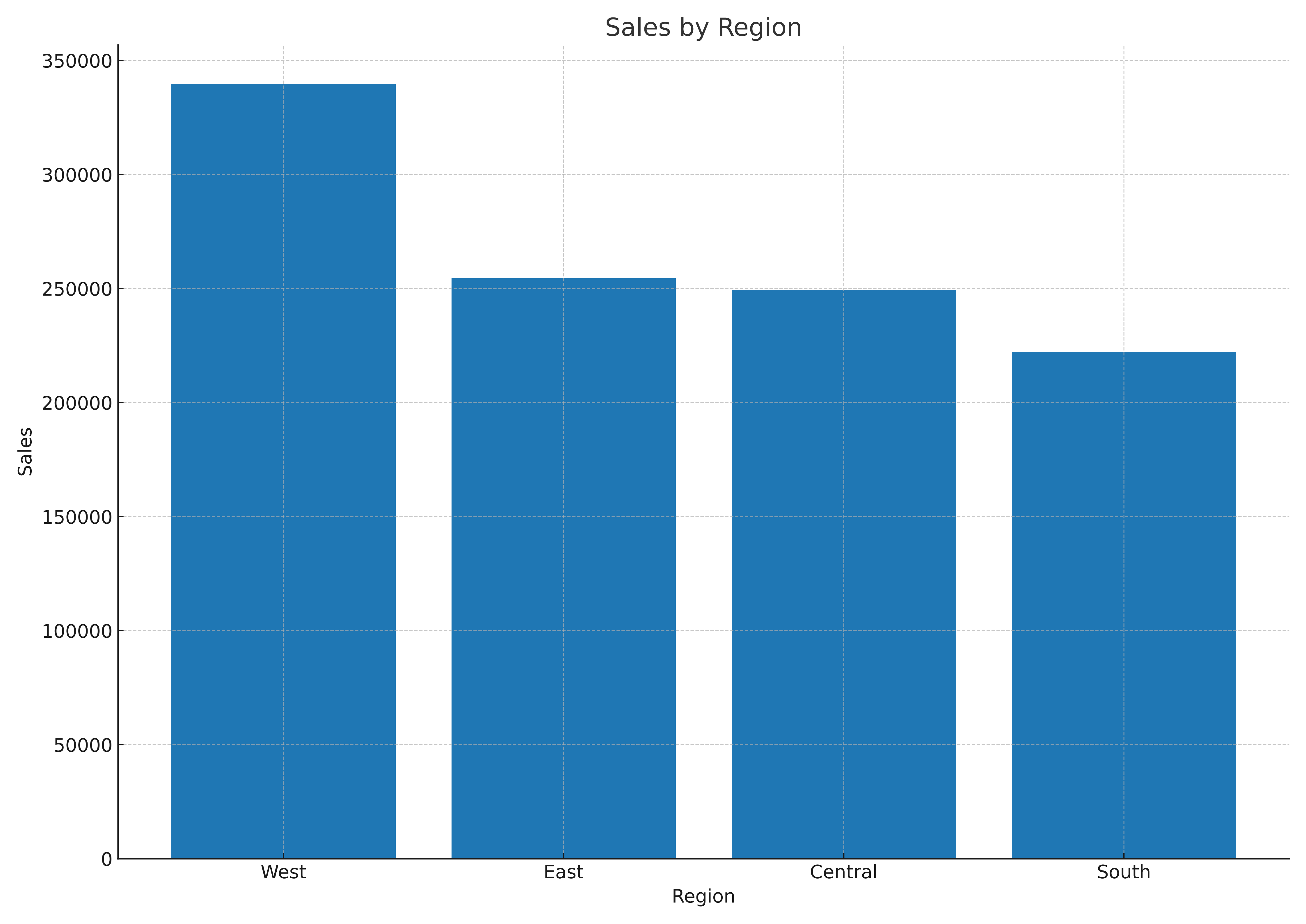
Python Code Listing (reproducible)

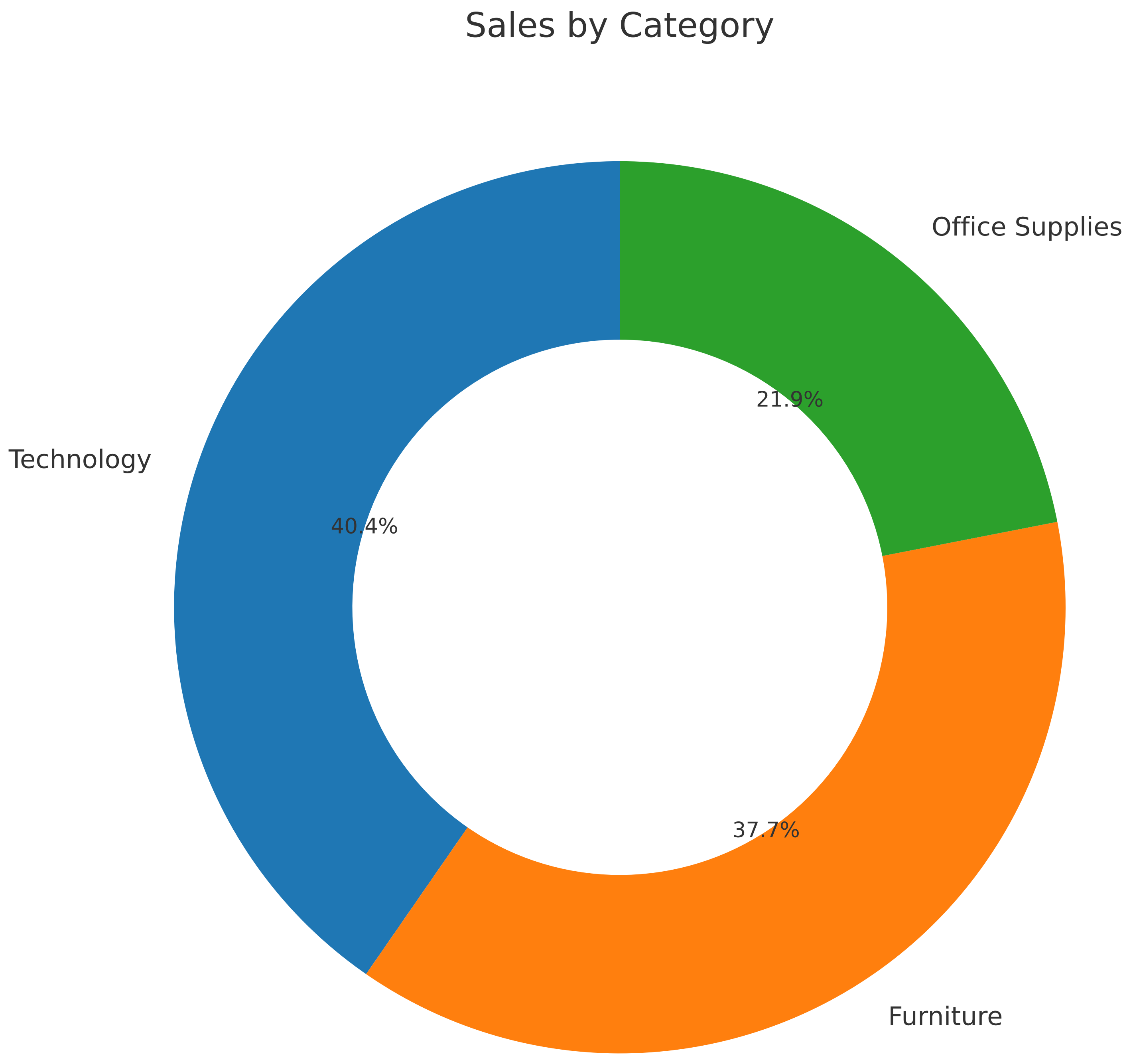
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from matplotlib.backends.backend_pdf import PdfPages
from datetime import datetime

# Load data
df = pd.read_csv("Superstore_Sales_Synthetic.csv", parse_dates=["Order Date"])
df["Month-Year"] = df["Order Date"].dt.to_period("M").astype(str)

# Aggregations
monthly_sales = df.groupby("Month-Year", as_index=False)["Sales"].sum()
monthly_sales["Month-Year"] = pd.to_datetime(monthly_sales["Month-Year"])
monthly_sales = monthly_sales.sort_values("Month-Year")

region_sales = df.groupby("Region", as_index=False)["Sales"].sum().sort_values("Sales", ascending=False)
category_sales = df.groupby("Category", as_index=False)["Sales"].sum().sort_values("Sales", ascending=False)

# Export charts to a PDF
with PdfPages("Recreated_Dashboard.pdf") as pdf:
    # Line chart: Sales over Months
    plt.figure(figsize=(10,5))
    plt.plot(monthly_sales["Month-Year"], monthly_sales["Sales"], marker="o")
    plt.title("Sales over Months")
    plt.xlabel("Month-Year")
    plt.ylabel("Sales")
    plt.xticks(rotation=45, ha="right")
    plt.tight_layout()
    pdf.savefig()
    plt.close()

    # Bar chart: Sales by Region
    plt.figure(figsize=(8,5))
    plt.bar(region_sales["Region"], region_sales["Sales"])
    plt.title("Sales by Region")
    plt.xlabel("Region")
    plt.ylabel("Sales")
    plt.tight_layout()
    pdf.savefig()
    plt.close()

    # Donut chart: Sales by Category
    plt.figure(figsize=(6,6))
    values = category_sales["Sales"].values
    labels = category_sales["Category"].values
```

Python Code Listing (reproducible)

```
wedges, texts, autotexts = plt.pie(values, labels=labels, autopct="%1.1f%%", startangle=90, wedgeprops=dict(width=0.4))
plt.title("Sales by Category")
plt.tight_layout()
pdf.savefig()
plt.close()
```


Sales over Months
Sales
Month-Year
62000
60000
58000
56000
54000
52000
50000
48000
2024-01
2024-03
2024-05
2024-07
2024-09
2024-11
2025-01
2025-03
2025-05
2025-07


Sales by Region
Sales
350000
300000
250000
200000
150000
100000
50000
0
West
East
Central
South
Region


Sales by Category
Office Supplies
21.9%
Technology
40.4%
37.7%
Furniture

Insights

1) Top Region: West (Total Sales ≈ 339,783).

2) Leading Category: Technology (Total Sales ≈ 430,232).

3) Peak Month: May 2025 with Sales ≈ 62,223.

4) Recent Trend: Last 3 months vs prior 3 months ≈ 17.2% change.